

Lewis T. Kunik

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Education

2026	<i>Ph.D.</i>	University of Utah	Atmospheric Sciences
2016	<i>B.S.</i>	University of Wisconsin-Madison	Computer Engineering & Computer Science

Employment

2026 -	Postdoctoral Researcher	University of Utah Atmospheric Sciences
2021-2026	Graduate Research Assistant	University of Utah Atmospheric Sciences
2019-2021	Air Quality Consultant	Ramboll U.S. Corporation

Research and Professional Interests

- Applications of terrestrial remote sensing for mapping vegetation dynamics
- Large-dataset analytics for monitoring continental-scale land surface processes
- Climate change and human impacts on terrestrial ecosystems and biogeochemical cycles
- Understanding impacts of wildfire, drought, pest/pathogen using geostatistics and remote sensing
- Air quality monitoring in urban & natural environments using advanced computing techniques

Publications

L. Kunik, Bowling, D.R., Raczka, B., Hicke, J.A., Frankenberg, C., Cheng, R., Slaton, M.R., and Lin, J.C., 2026. Characterizing effects of tree mortality from wildfire and bark beetles using satellite observations of solar-induced chlorophyll fluorescence. *Remote Sens. Environ.*, 344, 15550. DOI:

<https://doi.org/10.1016/j.rse.2026.115550>

Media coverage: University of Utah [@theU Research Spotlight](#)

- Roten, D.R., Lin, J.C., **Kunik, L.**, Mallia, D.V., Wu, D., Oda, T., and Kort, E. The Information Content of Dense Carbon Dioxide Measurements from Space: An Urban-Focused Inversion Approach with Synthetic Data from the OCO-3 Instrument. *ESS Open Archive*. August 28, 2024. DOI: [10.22541/essoar.172485642.21372186/v1](https://doi.org/10.22541/essoar.172485642.21372186/v1)
- J.C. Yang, Bowling, D.R., Smith, K.R., **Kunik, L.**, Raczka, B., Anderegg, W.R.L., Bahn, M., Blanken, P., Richardson, A.D., Burns, S.P., Bohrer, G., Desai, A.R., Altaf Arain, M., Staebler, R.M., Ouimette, P.A., Munger, J.W. and Litvak, M.E., 2024. Forest carbon uptake as influenced by snowpack and length of photosynthesis season in seasonally snow-covered forests of North America. *Agric. For. Meteorol.*, 353, 110054. DOI: <https://doi.org/10.1016/j.agrformet.2024.110054>
- L. Kunik**, Bowling, D.R., Raczka, B., Frankenberg, C., Köhler, P., Cheng, R., Smith, K.R., Goulden, M., Jung, M. and Lin, J.C., 2023. Satellite-based solar-induced fluorescence tracks seasonal and elevational patterns of photosynthesis in California's Sierra Nevada mountains. *Environ. Res. Lett.*, 19(1), 014008. DOI: <http://doi.org/10.1088/1748-9326/ad07b4>
- D. V. Mallia, Mitchell, L.E., Gonzalez Vidal, A.E., Wu, D., **Kunik, L.**, and Lin, J.C. 2023. Can We Detect Urban-Scale CO₂ Emission Changes Within Medium-Sized Cities? *J. Geophys. Res. Atmos.*, 128, e2023JD038686. <https://doi.org/10.1029/2023JD038686>
- D.V. Mallia, Mitchell, L.E., **Kunik, L.**, Fasoli, B., Bares, R., Gurney, K., Mendoza, D., and Lin, J.C. 2020. Constraining urban CO₂ emissions using mobile observations from a light-rail public transit platform, *Environ. Sci. Technol.*, 54(24), p15613-15621. DOI: <http://doi.org/10.1021/acs.est.0c04388>
- L. Kunik**, Mallia, D.V., Gurney, K.R., Mendoza, D.L., Oda, T. and Lin, J.C., 2019. Bayesian inverse estimation of urban CO₂ emissions: Results from a synthetic data simulation over Salt Lake City, UT. *Elem. Sci. Anth.*, 7(1), p.36. DOI: <http://doi.org/10.1525/elementa.375>

In Review:

- X. Huo, Raczka, B., **Kunik, L.**, Hicke, J.A., Frankenberg, C., Cheng, R., Slaton, M.R., and Lin, J.C. What is the value of assimilating multiple data streams for improving carbon uptake in the western United States? In Revision at *J. Geophys. Res. Biogeosci.*

Grants

2025	NASA Future Investigators in Earth and Space Science and Technology Leveraging remote sensing of photoprotective pigments to understand drought, mortality, and carbon dynamics in Western US forests	\$50k
2025	Wilkes Center for Climate Science & Policy Dissertation Writing Fellowship	\$30k
2023	FLUXNET Secondment Early-Career Program Exploring photosynthetic responses to warming and drying climate shifts in evergreen forest biomes using the semi-arid Yatir forest in Israel as a case study.	\$5k

2022 **National Science Foundation Graduate Research Fellowship**
Diagnosing drought-induced conifer stress in mountains using solar-induced chlorophyll
fluorescence from satellites

\$144k

Workshops and First-author Presentations

2026

A carotenoid-based vegetation index for tracking photosynthetic phenology and drought response over North America [Poster Presentation] *Wilkes Climate Summit*. Salt Lake City, UT.

2025

Monitoring photosynthetic responses of wildfire and bark beetle-induced tree mortality using satellite observations of solar-induced chlorophyll fluorescence (SIF) [Oral Presentation] *American Geophysical Union Fall Meeting*. New Orleans, LA.

Mapping conifer forest drought response with photoprotective pigment dynamics across North America using the MODIS Chlorophyll-Carotenoid Index (CCI) [Oral Presentation] *American Geophysical Union Fall Meeting*. New Orleans, LA.

2024

What is the utility of satellite-based solar-induced fluorescence (SIF) for predicting drought-induced forest mortality in the Western US? [Poster Presentation] *American Geophysical Union Fall Meeting Innovation Session*. Washington, DC.

2023

Satellite-based solar-induced fluorescence tracks seasonal and elevational patterns of photosynthesis in California's Sierra Nevada mountains. [Oral Presentation] *AmeriFlux Annual Meeting*. Garner, MA.

Early-career remote sensing workshop instructor. *AmeriFlux Annual Meeting*. Garner, MA.

Satellite-based solar-induced fluorescence tracks seasonal and elevational patterns of photosynthesis in California's Sierra Nevada mountains. [Poster Presentation] *American Geophysical Union Fall Meeting Innovation Session*. San Francisco, CA.

2022

Multi-scale Constraints Towards Understanding Conifer Forest Carbon Uptake in Complex Terrain within the Carbon Monitoring System-Mountains Project. [Poster Presentation] *American Geophysical Union Fall Meeting*. Chicago, IL.

Participant in Fluxcourse: two week early-career workshop focusing on foundations of land-atmosphere flux measurement, modeling and synthesis. *University of Colorado Mountain Research Station*. Niwot Ridge, CO.

2016

Multiscale Observational Constraints on CO₂ Flux Estimation using an Inverse Modeling Approach. [Poster Presentation] *University of Wisconsin – Madison Atmospheric, Oceanic and Space Sciences Poster Reception* Madison, WI.

2015

Multiscale Observational Constraints on CO₂ Fluxes. [Oral Presentation] *NOAA Office of Education Science and Education Symposium*. Silver Spring, MD.

Awards and Honors

Future Investigators in NASA Earth and Space Science and Technology (FINESST) Fellow	2025
Graduate Fellow – Wilkes Center for Climate Science & Policy, University of Utah	2025
FLUXNET Secondment Award Recipient	2023
National Science Foundation Graduate Research Fellow	2022
Distinction in Computer Sciences & Computer Engineering Majors - UW-Madison	2016
U.S. DOE Science Undergraduate Laboratory Internship (SULI) Scholar	2016
Best Oral Presentation Award – NOAA Student Science and Education Symposium	2015
Meyerhoff Undergraduate Excellence Award for Leadership, Service, and Scholarship	2015
NOAA Ernest F. Hollings Undergraduate Fellow	2014

Technical Skills

Statistical and Geospatial Data Analysis:

- Python: Jupyter, scipy, scikit-learn, xarray, xESMF, geopandas, cartopy, rasterio, rioxarray, dask
- R: tidyverse (dplyr, ggplot2, tidyr), sf, ncdf4, raster, sp, rgdal
- ArcGIS, QGIS: Raster and Vector data visualization

High Performance Computing & Data Management:

- Advanced Linux/Unix environment expertise including shell scripting
- SLURM job scheduling and monitoring
- parallel batch processing
- automated backup systems (cron jobs, rclone)
- large-scale dataset management across distributed storage systems (Google Drive, Box)
- Software development with AI tools: CoPilot, OpenAI, Claude
- Git version control and repository management

Extensive experience with Earth science data products, models and data archives including:

- Satellite remote sensing datasets: MODIS, VIIRS, SRTM, OCO-2/3 (NASA); Landsat (USGS); S5P-PAL, ESA-CCI (European Space Agency); GCOM-SGLI (JAXA)
- Climate reanalysis: SNODAS, GRIDMET, TerraClimate, ERA, nClimGrid, SPEIbase
- Land surface characteristics: NLCD, ESA CCI Land Cover, SRTM and NOAA ETOPO DEMs
- Biogeochemical model-data fusion products: FLUXCOM, MODIS GPP, CarbonTracker, Solar-Induced Fluorescence products (GOSIF, CSIF, SIF-ESDR, RTSIF)

- Proficient in data acquisition through USGS LPDAAC, NASA AppEARS and Google Earth Engine platforms
- Experience producing datasets using NetCDF Climate and Forecasting (CF) metadata standards
- Community Land Model-5.0 and Community Earth System Model-2: proficiency running and analyzing model output
- Atmospheric dispersion modeling (WRF-STILT, HYSPLIT, AERMOD)

Eddy covariance flux data processing: partitioning of ecosystem fluxes through REdDyProc R package and ONEFLUX processing routines

Advanced Statistical Methods: Bayesian inverse modeling, spatial bootstrapping, multivariate kernel density estimation, generalized linear/additive models (GLM/GAM), and spatiotemporal analysis and resampling techniques

Additional experience in programming and analysis with Excel, JavaScript, MATLAB, Android Studio, HTML, Java, C

Professional Experience

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|---|--------------|
| Postdoctoral Researcher, University of Utah, Salt Lake City, UT | 2026-present |
| <ul style="list-style-type: none"> • Developed and validated a gap-filled, gridded vegetation index dataset over N. America from NASA MODIS data. | |
| Graduate Research Assistant, University of Utah, Salt Lake City, UT | 2021-2026 |
| <ul style="list-style-type: none"> • Conducted research on regional-scale forest carbon cycling using satellite-based optical remote sensing. | |
| Air Quality Consultant, Ramboll Group, Salt Lake City, UT | 2019-2021 |
| <ul style="list-style-type: none"> • Performed comprehensive regulatory emissions and modeling analyses for industrial and commercial clients. | |
| Research Associate, Land-Atmosphere Interactions Research Group,
University of Utah, Salt Lake City, UT | 2017-2019 |
| <ul style="list-style-type: none"> • Applied Bayesian statistics to urban greenhouse gas data and models to constrain emissions and quantify uncertainties. | |
| Secondary Education Math Teacher, U.S. Peace Corps, Lesotho | 2018 |
| <ul style="list-style-type: none"> • Taught high school math in a rural district of Lesotho (sub-Saharan Africa) using English and Sesotho. | |
| Contract Researcher, NOAA Earth System Research Laboratory, Boulder, CO | 2017 |
| <ul style="list-style-type: none"> • Assessed network densities and meteorological drivers for data assimilation studies using aircraft and in-situ CO₂ data. | |
| Electrical Engineering Intern, National Renewable Energy Laboratory, Golden, CO | 2016 |
| <ul style="list-style-type: none"> • Developed a custom control board and solid-state relay configuration for residential “smart” appliances. | |

Research Assistant, Desai Ecometeorology Lab, Dept. of Atmospheric Sciences, University of Wisconsin, Madison, WI, 2015-2016

- Integrated ground-based spectrometer data into regional Bayesian inversion techniques to constrain upper tropospheric carbon fluxes.

Hollings Fellow, NOAA Earth System Research Laboratory, Boulder, CO 2015

- Performed synthetic data analyses of Bayesian statistical modeling techniques to aid in regional CO₂ flux model development.

Chainsaw Crew Member, Rocky Mountain Conservancy, Grand Lake, CO 2014

- Felled hazardous beetle kill trees near high-use recreation areas in response to Mountain Pine Beetle outbreak

Lab Member, Holloway Air Quality Research Group, University of Wisconsin, Madison, WI 2013

- Examined trends in air pollution and temperature extremes over cities in the US Midwest.

Teaching and Mentoring Experience

2024-present

Adult English Language Tutor, Guadalupe School, Salt Lake City, UT

2022

Software programming mentor, REALM– Research Experience for Undergraduates (REU) program, University of Utah, Salt Lake City, UT

2018

Instructor, U.S. Peace Corps Math Education Sector, Lesotho

2017

Instructor, Bayesian Inverse Modeling Workshop, University of Utah, Salt Lake City, UT

Instructor, Bayesian Inverse Modeling Workshop, NOAA Earth System Research Lab, Boulder, CO

Professional Service and Volunteerism

Manuscript referee: Global Change Biology, Science of the Total Environment, Geophysical Research Letters, Environmental Research Letters, Forest Ecology and Management Journals
IOP Trusted reviewer status

Software programming and English language mentor volunteer, Smithsonian Tropical Research Institute, Gamboa, Panama 2023

Mentor, City of South Salt Lake Promise Program, South Salt Lake, UT 2020

Mentor, Madison Middle School Science Symposium, Wright Middle School, Madison, WI 2016

Vice President and Project Manager, Engineers without Borders, University of Wisconsin,

Madison, WI	2015-2016
Education Outreach Coordinator, Engineers without Borders, University of Wisconsin, Madison, WI	2014
STEM Education Volunteer, Madison Children's Museum, Madison, WI	2014
Mentor, Science Olympiad, Edgewood High School, Madison, WI	2013 – 2014

Professional Organizations and Certifications

Member - American Geophysical Union	2022-present
Member - Air & Waste Management Association, Great Basin Chapter	2019-2020
EPA Method 9 Visible Emissions Certification (through AeroMet Engineering)	2020

Languages Spoken

English (native), Spanish (conversational)

Personal Interests and Accomplishments

Long distance hiking: completion of Superior Hiking Trail (2015), Pacific Crest Trail (2017), Arizona Trail (2018), Hayduke Trail (2021), Wind River High Route (2022) and Great Divide Trail (2026)

Cross country skiing: 12-time finisher of American Birkebeiner (Cable, WI)

Curling Instructor at the Utah Olympic Oval, Kearns, UT (2018 – 2022)

Backcountry skiing, canoeing, mountain biking, trail running, climbing, bicycle touring, canyoneering, classical piano